INCH-POUND

MIL-S-81282B
22 January 1993
Superseding
MIL-S-81282A
6 December 1973

MILITARY SPECIFICATION
STANDARD, FREE WATER DETECTOR

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This Specification covers the comparator standard to be used with the Free Water Detector conforming to MIL-D-81227.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

PPP-T-60	Tape: Packaging Waterproof
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-601	Box, Wood Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-D-81227	Detector Kit, Water, Automotive-Aviation Fuel
MIL-D-81248	Detector Pad, Free Water

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Air Systems Command, AIR-5363, Washington, DC 20361-5360, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military standards are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall conform to applicable specifications and shall be specified herein. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality suitable for the purpose intended.

3.1.1 Paper. The chemical composition of the paper shall be in accordance with type ORO-T4 manufactured by Color Crester Co., 9707 Emily Rd., Levittown, PA 19057, or approved equivalent. The paper shall have a pressure-sensitive backing consisting of a water insoluble adhesive with carrier or separator sheet applied over the adhesive.

3.1.2 Inks.

3.1.2.1 Formula A. Formula A ink shall consist of 100 percent black ink.

3.1.2.2 Formula B. Formula B ink shall consist of 15.0 percent (all formulation percentages by weight) IRB-17 Yellow, 40.0 percent YA 8500 ATLAS Yellow, 25 percent 9PI White, 9 percent 902 XL Compound, 8 percent M52 Oil, 2 percent Tri Decyl Alcohol, 0.50 percent of 6 percent Cobalt Drier, and 0.50 percent of 6 percent Manganese Drier. Formula B ink is available from Rexford Co., 9th & Sring Garden Streets, Philadelphia, PA 19123. Equivalent ingredients and suppliers maybe used provided approval is obtained from the Naval Air Warfare Center, Aircraft Division, PE33, P.O. Box 7176, Trenton NJ 08628-0176 (NAWCADTRN).

3.1.2.3 Formula C. Formula C shall consist of 80.5 percent (all formulation percentages by weight) Visiprint VF, 10.5 percent RV 687 Gel Varnish, 3.5 percent Magiesol 47 Oil, 0.80 percent of 6 percent Manganese Drier, 0.60 percent of 18 percent Zirconium Drier, 3.1 percent Polyethylene Wax Compound, and 1.0 percent Teflon Wax Compound and is available from Radiant Color Co., 2800 Radiant Ave., PO Box 4019 Richmond, CA 94804-0019. Equivalent ingredients and suppliers maybe used provided approval is obtained from NAWCADTRN.

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3.1.3 Engraved plates. A set of engraved printing plates shall be supplied by NAWCADTRN to the contractor when a contract is awarded. The plates shall be designated 1, 2, and 3 (see 3.3). The contractor shall make molds of the plates, to be used in the printing process to avoid damage to the original plates.

3.2 Design. The standard shall be a comparator capable of determining the amount of free water in the aviation fuel when a detector pad, in accordance with MIL-D-81248, is compared with the applicable patch on the standard. The design of the standard shall conform to MIL-D-81227.

3.3 Construction. The Free Water Detector Standards shall be produced in accordance with the steps specified in paragraphs 3.3.1 through 3.3.4 below using the printing plates produced in accordance with paragraph 3.1.3.

3.3.1 Step 1. Plate 1 (3.1.3) and formula A (3.1.2.1) ink shall be used to produce a black overprint for the visible layout of the Free Water Detector standard.

3.3.2 Step 2. Plate 2 (3.1.3) and formula B ink (3.1.2.2) shall be used to produce a background fluorescent pattern for patches 5, 10, and 20 on the Free Water Detector Standard.

3.3.3 Step 3. Plate 3 (3.1.3) and formula C ink (3.1.2.3) shall be used to produce an overall fluorescent pattern for patches 5, 10, and 20 on the Free Water Detector Standard.

3.3.4 Dimensions. The trim sizes of the Free Water Detector Standard shall be 1¾ by 9 inches. The tolerance shall be ± 0.125 inch.

3.4 Fluorescence Performance. Each standard shall meet the fluorescence requirements of Table I when tested in accordance with paragraph 4.6 at the conclusion of manufacturing steps 2 and 3 (see paragraphs 3.3.1 and 3.3.2). Requirements contained in Table I are absolute and shall not be subjected to correction for test tolerances.

TABLE I
FREE WATER DETECTOR STANDARD FLUORESCENT REQUIREMENTS

FREE WATER DETECTOR STANDARD	STEP 2		STEP 3	
	Minimum	Maximum	Minimum	Maximum
5 PPM	6.7	9.5	10.4	13.6
10 PPM	11.0	15.8	20.8	28.0
20 PPM	16.9	29.7	47.4	56.2

3.5 Storage. Finished standards when packed in accordance with 5.2 shall be placed in a humidity controlled storage area.

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3.6 Workmanship. The standard shall be so constructed and finished that it will be free, when examined visually, of any defects which may affect serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as quality conformance inspection.

4.3. Article inspection. All sampling shall conform to MIL-STD-105. The minimum inspection level shall not be less than one article out of every 200 articles produced. One comparator standard constitutes one article. Article inspection shall include all the examinations and tests of this specification (4.4 and 4.6) except the inspection of packaging (4.7). Unless otherwise specified, no material shall be submitted for acceptance under any contract or order until the article samples prescribed in 4.3.1 have been subjected to article inspection and pronounced satisfactory by the procuring activity. However, approval of the article samples or equivalent certification and data shall not relieve the supplier of his obligation to meet the quality conformance inspection.

4.3.1 Selection of article samples. Unless otherwise specified in the contract or purchase order, the random samples for article inspection shall be selected from each inspection lot (see 4.3.1.1) in accordance with MIL-STD-105. The unit for determining lot size and the sample unit shall be as specified in the applicable examination or test paragraph and the inspection level.

4.3.1.1 Inspection lot. A inspection lot shall consist of Free Water Detector Standards that have been printed at essentially the same time under the same conditions. The Sample unit shall be one fully prepared shipping container ready for delivery with the exception that it need not be sealed.

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4.3.2 Inspection Conditions. Inspections shall be conducted at ambient conditions.

4.4 Examinations.

4.4.1 Examination of Free Water Detector Standards. When visually examined each Free Water Detector Standard shall be evenly coated and free from foreign matter, holes, tears, cuts, greases, wrinkles or other imperfections which would render the standard unsuitable for its intended use. Each standard shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship and marking.

4.4.2 Examination for registration. Each Free Water Detector Standard shall be tested visually under ultraviolet light for registration, using a Free Water Detector conforming to MIL-D-81227.

4.4.3 Examination for fluorescence. A Fluorometer shall be used to monitor the fluorescence of the standard at the conclusion of printing steps 2 and 3 (see 3.3.2 and 3.3.3). Each Free Water Detector Standard shall conform to the appropriate minimum and maximum requirements of Table I when tested in accordance with paragraph 4.6.

4.5 Rejection. Failure of any sample of the Free Water Detector Standards to conform to any one of the requirements to this specification shall be cause for rejection of the lot represented. Rejected material shall not be resubmitted without furnishing full particulars concerning previous rejection and measures taken to overcome defects.

4.6 Test methods.

4.6.1 Apparatus. The apparatus shall consist of a Turner Fluorometer 111 used to monitor the fluorescent in 3.3.2 and 3.3.3. The Fluorometer shall have a paper chromatogram door accessory. This door enables a sample standard to be scanned in order to assess its compliance with Table I.

4.6.2 Fluorometer 111. The meter shall be equipped with a general purpose ultra violet lamp (emission between 310 nm and 390 nm), emission filter 2A-12, excitation filter 47B, and a neutral density filter having a nominal range of 1%. The instrument sensitivity shall be set at 10X.

4.7 Packaging inspection. MIL-STD-105 inspection level S-2 shall be imposed on the fully prepared shipping containers. Each of the fully prepared shipping containers selected as a sample unit shall be visually examined to determine that preservation, packing and marking conform to requirements of Section 5.

5. PACKAGING

5.1 Preservation.

5.1.1 Unit container. Each standard shall be individually packaged in a light-proof, moisture-proof envelope.

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5.1.2 Intermediate packaging. The standards, packaged as specified in 5.1.1, shall be further packaged in multiples of 50 in intermediate containers conforming to PPP-B-566.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Intermediate containers shall be packed in a close fitting shipping container constructed, closed and strapped in accordance with class weather-resistant compliance symbol V3C or V3S, Style RCS of PPP-B-636. Net weight of contents in each shipping container shall not exceed 65 pounds. Each fiberboard shipping container shall be further sealed with a 3 inch minimum width pressure sensitive tape conforming to Class III of PPP-T-60. The tape shall be evenly applied over the full length of all seams and closures including the manufacturer's joint and extending at least three inches onto each end and side panels as applicable. When specified, (see 6.2) for Army use, the intermediate containers shall be packed in wood-cleated plywood containers conforming to PPP-B-601, Grade A, closed in accordance with the Appendix thereto and subject to the weight limitations of PPP-B-601.

5.2.2 Level B. Intermediate containers shall be packed in a close fitting container constructed in accordance with PPP-B-636, class domestic, Style RSC. Net weight of contents in each shipping container shall not exceed 65 pounds. All flaps of the shipping container shall be securely sealed with an adhesive used commercially for sealing fiberboard boxes by application throughout the entire area of contact between the flaps or by a combination of metal stitching the bottom and sealing the top flaps with adhesive.

5.2.3 Commercial. Packages that require overpacking for acceptance by the common carrier shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest cost to the point of delivery.

5.3 Marking. Each container, including the unit container, shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking. Each packaging envelope shall be marked with the following:

(a) National Stock Number (NSN 6640-00-999-2784)

(b) STANDARD, FREE WATER DETECTOR

(c) 1 EACH MIL-S-81282B

(d) Date of printing

(e) Expiration date (3 years from the date of printing).

(f) Contract No., Lot No.

(g) Printer's name, address

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(h) CAUTION: Improper handling or continual exposure to sunlight, chemical fumes, etc. may cause colors of the standard to fade. User should exercise care to insure adequate protection to preclude color change.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The standard covered by this specification is intended for use with Free Water Detector conforming to MIL-D-81227. The standard shall be used as a comparator to determine the degree of water contaminant in aircraft fuel when the Detector Pad (MIL-D-81248) is compared with the standard under ultraviolet light.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Quantity

d. Level of packing (see 5.2)

6.3 Special notice. The Free Water Detector Standards covered by this specification are not satisfactorily reproducible by the offset process. Satisfactory reproduction can be made only by using the duplicate plates or molds (3.1.3) which are made from the original plates supplied by NAWCADTRN.

6.4 The Naval Air Warfare Center, Aircraft Divison, (NAWCADTRN) PE33, P.O. Box 7176, Trenton, NJ 08628-0176 is the activity that is responsible for deciding whether alternate materials, that are offered, are the equivalent of those specified herein. However, bidders should submit detailed information of their offerings through the procurement officer of the cognizant procuring agency for a determination as to acceptability.

6.5 Subject term (key word) listing.

Free Water Detector Standard
Fluorescence
Fluorometer
Free Water Detector Pad
Engraved plates

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army - MR
Air Force - 68

User Interest:
Army - ME

Preparing activity:
Navy - AS
(Project No. 6640-1797)

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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-81282B	2. DOCUMENT DATE (YYMMDD) 930122

3. DOCUMENT TITLE
STANDARD, FREE WATER DETECTOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Naval Air Systems Command AIR-5363C	b. TELEPHONE (Include Area Code) (1) Commercial 202-692-2653 (2) AUTOVON 222-2653
c. ADDRESS (Include Zip Code) Naval Air Systems Command AIR-5363C Washington, DC 20361-5360	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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